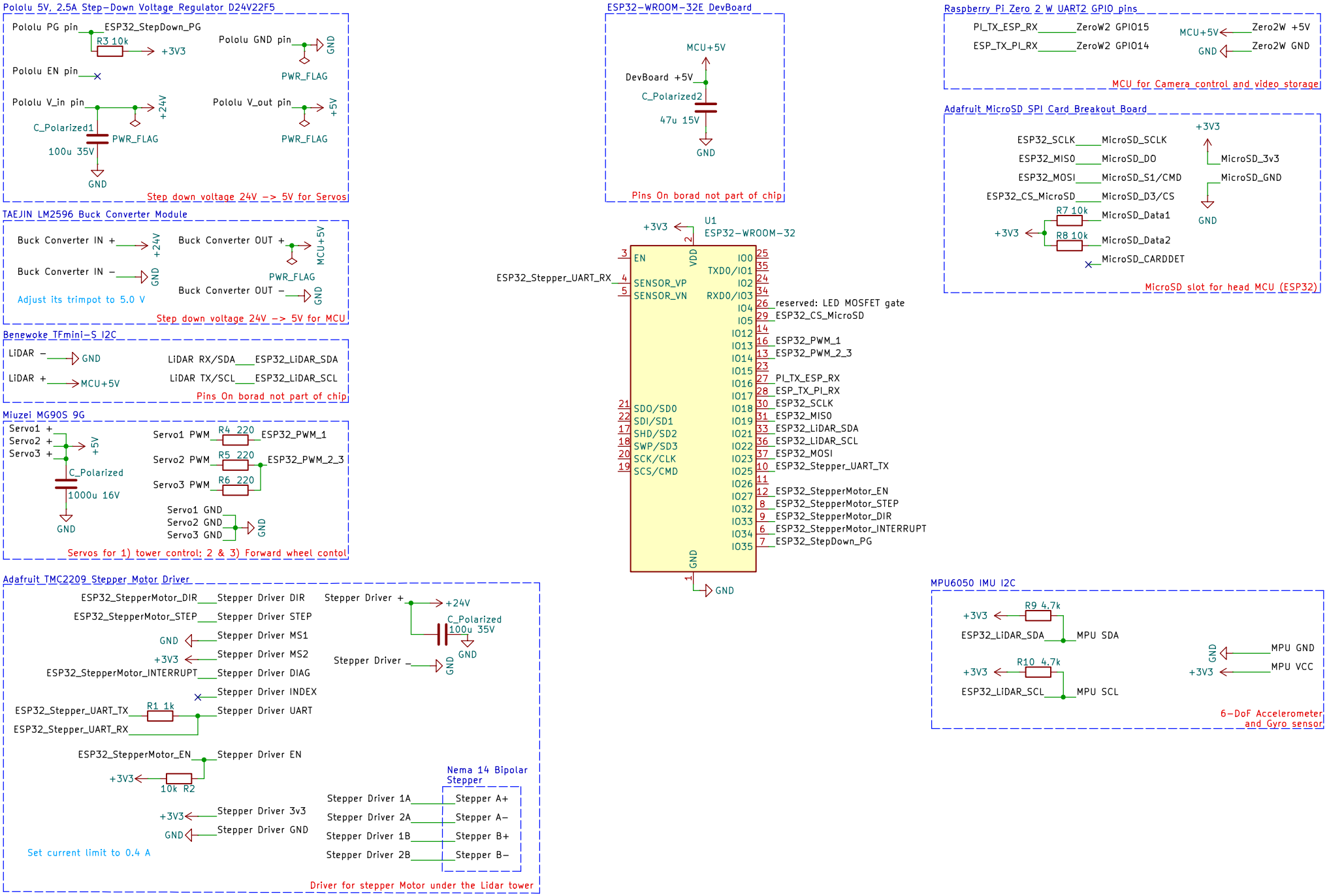

Pololu 5V, 2.5A Step-Down Voltage Regulator D24V22F5
Pololu PG pin
ESP32_StepDown_PG
R3 10k
+3V3
Pololu GND pin
GND
PWR_FLAG
Pololu EN pin
Pololu V_in pin
+24V
C_Polarized1
PWR_FLAG
100u 35V
GND
Pololu V_out pin
+5V
PWR_FLAG
Step down voltage 24V -> 5V for Servos
TAEJIN LM2596 Buck Converter Module
Buck Converter IN +
+24V
Buck Converter IN -
GND
Adjust its trimpot to 5.0 V
Buck Converter OUT +
MCU+5V
PWR_FLAG
Buck Converter OUT -
GND
Step down voltage 24V -> 5V for MCU
Benewoke TFmini-S I2C
LiDAR -
GND
LiDAR +
MCU+5V
LiDAR RX/SDA
ESP32_LiDAR_SDA
LiDAR TX/SCL
ESP32_LiDAR_SCL
Pins On borad not part of chip
Miuzei MG90S 9G
Servo1 +
Servo2 +
Servo3 +
+5V
C_Polarized
1000u 16V
GND
Servo1 PWM
R4 220
ESP32_PWM_1
Servo2 PWM
R5 220
ESP32_PWM_2_3
Servo3 PWM
R6 220
Servo1 GND
Servo2 GND
Servo3 GND
GND
Servos for 1) tower control: 2 & 3) Forward wheel contol
Adafruit TMC2209 Stepper Motor Driver
ESP32_StepperMotor_DIR
Stepper Driver DIR
ESP32_StepperMotor_STEP
Stepper Driver STEP
GND
Stepper Driver MS1
+3V3
Stepper Driver MS2
ESP32_StepperMotor_INTERRUPT
Stepper Driver DIAG
Stepper Driver INDEX
ESP32_Stepper_UART_TX
R1 1k
Stepper Driver UART
ESP32_Stepper_UART_RX
ESP32_StepperMotor_EN
Stepper Driver EN
+3V3
10k R2
+3V3
Stepper Driver 3v3
GND
Stepper Driver GND
Set current limit to 0.4 A
Stepper Driver +
+24V
C_Polarized
100u 35V
GND
Stepper Driver -
GND
Nema 14 Bipolar Stepper
Stepper Driver 1A
Stepper A+
Stepper Driver 2A
Stepper A-
Stepper Driver 1B
Stepper B+
Stepper Driver 2B
Stepper B-
Driver for stepper Motor under the Lidar tower
ESP32-WROOM-32E DevBoard
MCU+5V
DevBoard +5V
C_Polarized2
47u 15V
GND
Pins On borad not part of chip
+3V3
U1
ESP32-WROOM-32
EN
VDD
SENSOR_VP
SENSOR_VN
ESP32_Stepper_UART_RX
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
GND
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
reserved: LED MOSFET gate
IO5
ESP32_CS_MicroSD
IO12
IO13
ESP32_PWM_1
ESP32_PWM_2_3
IO14
IO15
IO16
PI_TX_ESP_RX
IO17
ESP_TX_PI_RX
IO18
ESP32_SCLK
IO19
ESP32_MISO
IO21
ESP32_LiDAR_SDA
IO22
ESP32_LiDAR_SCL
IO23
ESP32_MOSI
IO25
ESP32_Stepper_UART_TX
IO26
IO27
ESP32_StepperMotor_EN
IO32
ESP32_StepperMotor_STEP
IO33
ESP32_StepperMotor_DIR
IO34
ESP32_StepperMotor_INTERRUPT
IO35
ESP32_StepDown_PG
GND
Raspberry Pi Zero 2 W UART2 GPIO pins
PI_TX_ESP_RX
ZeroW2 GPIO15
ESP_TX_PI_RX
ZeroW2 GPIO14
MCU+5V
Zero2W +5V
GND
Zero2W GND
MCU for Camera control and video storage
Adafruit MicroSD SPI Card Breakout Board
+3V3
ESP32_SCLK
MicroSD_SCLK
ESP32_MISO
MicroSD_DO
ESP32_MOSI
MicroSD_S1/CMD
ESP32_CS_MicroSD
MicroSD_D3/CS
MicroSD_3v3
MicroSD_GND
GND
R7 10k
+3V3
R8 10k
MicroSD_Data1
MicroSD_Data2
MicroSD_CARDDET
MicroSD slot for head MCU (ESP32)
MPU6050 IMU I2C
R9 4.7k
+3V3
ESP32_LiDAR_SDA
MPU SDA
R10 4.7k
+3V3
ESP32_LiDAR_SCL
MPU SCL
GND
MPU GND
+3V3
MPU VCC
6-DoF Accelerometer and Gyro sensor